

Pablo Leyva

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EDUCATION

New Jersey Institute of Technology

B.S. in Applied Mathematics (Statistics) & Computer Science (AI Concentration)

Newark, NJ

Aug 2023 – May 2027

- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Database Systems, Machine Learning, Deep Learning, Reinforcement Learning, Probability, Linear Algebra, Convex Optimization, Casual Inference, Natural Language Processing, Computer Vision, Real Analysis, Stochastic Processes, Time Series Analysis

EXPERIENCE

Apple

AI/ML Intern

May 2026 – Present

Cupertino, CA

- Researched 6+ use cases of **SIRI agentic reasoning** within the 18 wallet products with a product focus of building trust between user and AI, demonstrating the application of **UCP** to help for a new set of usecases for SIRI, Wallet and Commerece
- Built a demo using custom **agent harness** that **predicts user needs** by using onscreen information from messages coupled with Event Pass information that serves as grounding information from user's wallet
- Cross **collaborated with 4 teams** for engineering, product, marketing, and research to synthesis technical, product, researcha and business views and requirements to build demo aligned with vision and goals

Undergrad Research and Innovation (URI) - Agentic Reasoning SLM

Research Fellow

Sep 2025 – Present

Newark, NJ

- Finetuned a **3B-parameter reasoning SLM** via **SFT** → **DPO** → **RL** with multi-GPU distributed training (FSDP) on math-proof and logic-puzzle datasets, studying one-step reasoning generalization in small language models
- Constructed a **synthetic RL environment** with programmatically-checkable rewards for mathematical proof search; ran **ablation studies** across reward-shaping strategies and data-mixing ratios to isolate sources of reasoning gain
- Ran **linear probes** on hidden activations and **safety evaluations** (refusal consistency, reward hacking, out-of-distribution robustness), benchmarking the SLM against larger LLM baselines on lm-eval-harness

Apple

AI/ML Intern

May 2025 – August 2025

Cupertino, CA

- Built an **RAG agent orchestration system** integrating the Apple Foundation Model (AFM) with Model Context Protocol (MCP); designed stateful multi-step tool execution with retry and human-in-the-loop fallback paths
- Built the team's **LLM evaluation pipeline** benchmarking agent trajectories (tool-use success, groundedness, hallucination rate)
- Designed **tool selection and execution** flows with LangGraph to collect **trajectory-level preference data** across entire Apple Pay Technical Docs; presented the prototype at the team's end-of-internship review (org director present)

Caterpillar Inc.

Software Engineering Intern

May 2024 – August 2024

Chicago, IL

- Built an **end-to-end data pipeline** with Python and Apache Airflow (scheduled DAGs with retry and failure handling) ingesting 500K+ daily records; served model predictions via **FastAPI microservices** at sub-100ms latency for 2K+ daily requests
- Fine-tuned **transformer LLMs** (BERT, GPT-3.5) with **LoRA/QLoRA** on a 500K-record corpus; ran ablation studies on data mixing and learning-rate schedules, reaching 89% classification accuracy

PROJECTS

Production-Grade Personalized Recommendation System | *PyTorch, XGBoostPostgreSQL, Docker, MLflow, AWS/GCP* | **Python**

- Trained user and item embeddings with PyTorch, implemented **negative sampling** and cold-start strategies for limited interaction
- Developed a ranking pipeline using **XGBoost**, incorporating behavioral, similarity, and contextual features.
- Deployed the recommendation pipeline as a **FastAPI REST service** with PostgreSQL-backed features and Dockerized inference.

Self-Evolving Agent Harness | *PyTorch, TRL, Gymnasium, vLLM, LoRA, Weights & Biases* | **Python**

- Built the first **agent harness for non-technical users** that auto-generates **personalized tools** from natural-language needs and curates on-policy trajectories to **fine-tune a per-user policy (LoRA)**, internalizing recurring context so the agent no longer re-reads static memory files, cutting per-turn tokens by ~60%

Multimodal Product Search & Cross-Modal Retrieval System | *PyTorch, CLIP, Hugging Face, FAISS, Transformers, FastAPI, Docker*

- Built a multimodal product search and retrieval system that combined text and image embeddings for semantic catalog search and recommendation support.

COMMUNITY OUTREACH

President, Vice President - External Affairs, Event Coordinator

Society of Hispanic Professional Engineers (SHPE)

October 2023 – Present

Newark, NJ

- Leading a team of 20 to organize professional events for 300+ students; grew external company partnerships 6x and averaged 200% growth in weekly member engagement
- Built and shipped the chapter's first **iOS app** in 27 days with a team of 5 engineers, growing it to **200+ users** in its first month

TECHNICAL SKILLS

- Languages** : Python, Swift, C++, JavaScript/TypeScript, Java
- Tools & Infra** : Git, Docker, AWS (EC2, S3, Lambda), PostgreSQL, Linux
- ML Frameworks** : PyTorch, scikit-learn, JAX, Hugging Face Transformers, TRL
- Reinforcement Learning** : PPO, GRPO, DPO, RLHF, reward modeling, Gymnasium, RL environment design, preference data
- LLM Training & Evaluation** : SFT, LoRA/QLoRA, fine-tuning, distillation, synthetic data, lm-eval-harness, safety evaluations, ablation studies